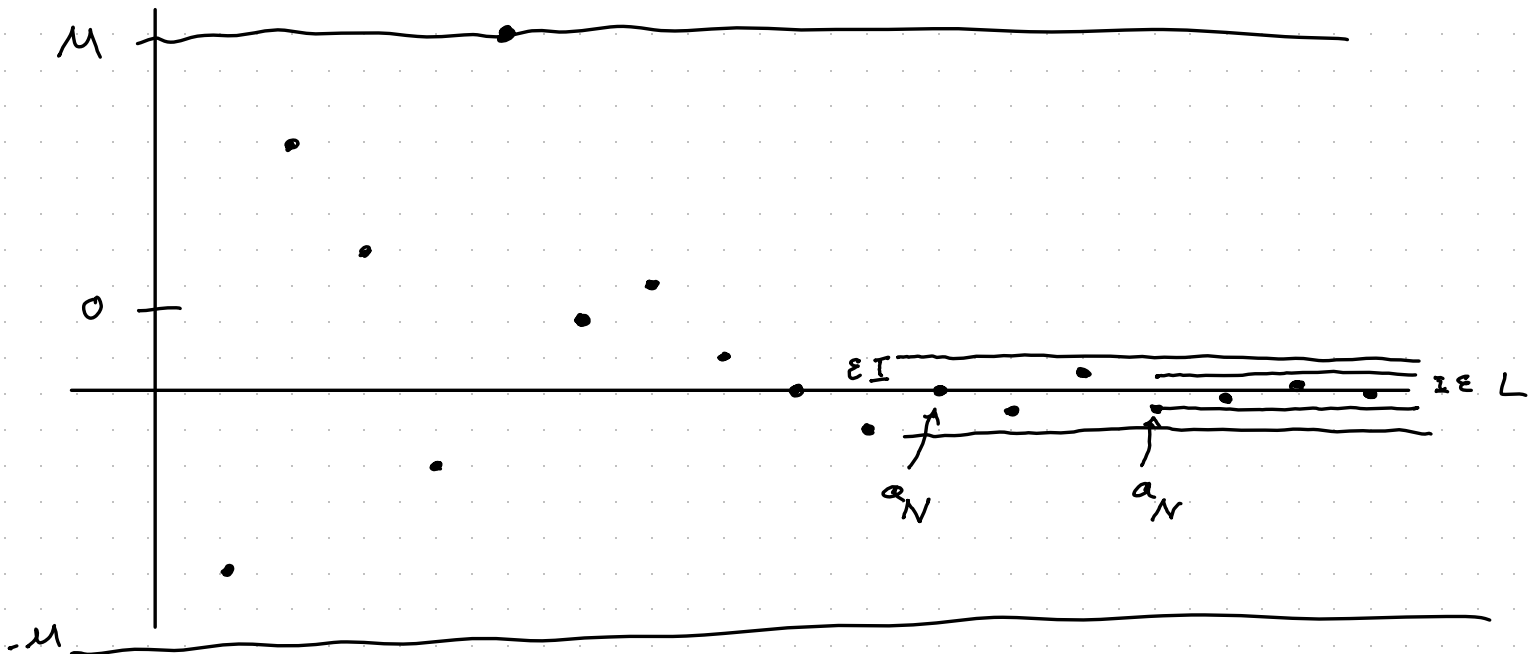
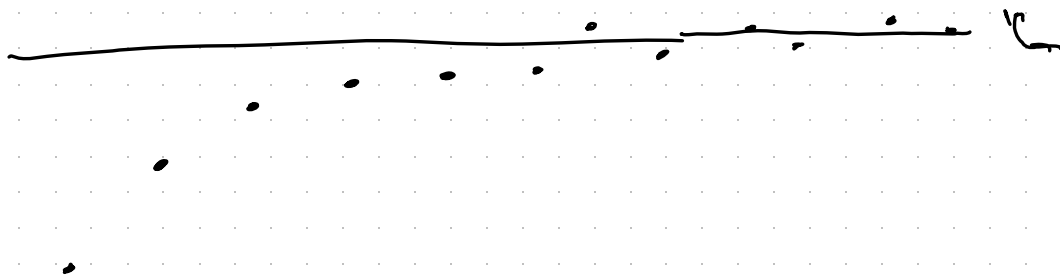


Formal definition of a limit — 5.1



Definition: We say (a_n) converges to L if for every $\epsilon > 0$, there exists an N such that $|a_n - L| < \epsilon$ for all $n \geq N$.



Proof that a convergent sequence is bounded:

Take N large enough that $|a_n - L| < 1$

for all $n \geq N$.

Then define $M = \max \{|a_1|, |a_2|, \dots, |a_{N-1}|, |L| + 1\}$.

Need to show that $|a_n| \leq M$ for all n .

$|a_n| \leq M$ for $n = 1, 2, \dots, N-1$.

If $n \geq N$, then $|a_n - L| < 1$, meaning

$$-1 < a_n - L < 1 \iff L-1 < a_n < L+1.$$

This means that $-|L|-1 \leq L-1 < a_n < L+1 \leq |L|+1$

$$-(|L|+1) < a_n < |L|+1$$

$$\Rightarrow |a_n| < |L|+1 \leq M$$

So $|a_n| \leq M$ in all cases, and this

means the sequence (a_n) is bounded. $\square$